

14.1 Random regressors and conditional inference

The theory of Part III was built for a model matrix $\mathbf{X}$ of known constants. In observational data the regressors are drawn with the response, and a second sample would bring a different $\mathbf{X}$. Section 5.1 already argued that the assumptions of the linear model should then be read *conditionally on $\mathbf{X}$* , and Corollary 5.5.5 carried unbiasedness over to random regressors. This section does the same for the exact distribution theory. Tests and intervals stay exact; the estimates acquire heavier-tailed distributions, and power becomes a random quantity that depends on the design the data deliver.

14.1.1 The model with random regressors

Definition 14.1.1 — Normal linear model with random regressors

Let $\mathbf{X}$ be a random $n \times p$ matrix and $\mathbf{Y}$ a random n -vector. They follow the **normal linear model with random regressors** if $\text{rank}(\mathbf{X}) = p$ with probability one and, conditionally on $\mathbf{X}$,

$$\mathbf{Y} \mid \mathbf{X} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n)$$

for constants $\boldsymbol{\beta} \in \mathbb{R}^p$ and $\sigma^2 > 0$. Equivalently, $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ independent of $\mathbf{X}$.

The leading case has independent and identically distributed rows $(\mathbf{x}_{(i)}^\top, Y_i)$ with $Y_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \varepsilon_i$, where $\varepsilon_i \sim N(0, \sigma^2)$ is independent of $\mathbf{x}_{(i)}$. Independence of the rows then makes the errors independent given the whole of $\mathbf{X}$. Jointly normal rows satisfy the definition (Eq. 3.5.2), but the regressors may equally be skewed, discrete or dependent on one another; only the errors must be normal, homoscedastic and independent of the regressors.

14.1.2 Conditioning on the design

Lemma 14.1.1 — Pivots given the design

Let S be a random vector and U any random element. Suppose there is a distribution Q such that, for almost every value u , the conditional distribution of S given $U = u$ is Q . Then $S \sim Q$, and S is independent of U .

Proof. For a Borel set B and an event $\{U \in A\}$, the tower rule gives

$$\begin{aligned} \Pr(S \in B, U \in A) &= E[\mathbf{1}\{U \in A\} \Pr(S \in B \mid U)] \\ &= E[\mathbf{1}\{U \in A\} Q(B)] = Q(B) \Pr(U \in A). \end{aligned}$$

Taking A to be the whole space gives $\Pr(S \in B) = Q(B)$, and then the display says that the joint probability factorizes, which is independence.

A statistic whose conditional law given $\mathbf{X}$ is the same for every design is a *pivot given the design*. The t and F statistics of Part III are of this kind.

Theorem 14.1.2 — Conditional inference is valid unconditionally

Assume the model of Definition 14.1.1, with $n > p$. Let $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ and $s^2 = \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 / (n - p)$.

- (a) For every fixed $\boldsymbol{\lambda} \neq \mathbf{0}$, the statistic $T = (\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta}) / (s \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}})$ has the $t(n - p)$ distribution and is independent of $\mathbf{X}$.
- (b) Let $\boldsymbol{\Lambda}$ be a fixed $p \times q$ matrix of rank q , let $\mathbf{d} \in \mathbb{R}^q$, and put

$$F = \frac{(\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top [\boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\Lambda}]^{-1} (\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})}{q s^2}.$$

Given $\mathbf{X}$, $F \sim F(q, n - p, \gamma(\mathbf{X}))$ with

$$\gamma(\mathbf{X}) = \frac{(\boldsymbol{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})^\top [\boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\Lambda}]^{-1} (\boldsymbol{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})}{\sigma^2}. \quad (14.1.1)$$

If $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$, then $F \sim F(q, n - p)$ unconditionally, independently of $\mathbf{X}$.

- (c) A confidence set whose coverage probability given $\mathbf{X}$ equals $1 - \alpha$ for almost every $\mathbf{X}$ has unconditional coverage $1 - \alpha$. A test whose rejection probability under the null hypothesis equals α given almost every $\mathbf{X}$ has unconditional size α .
- (d) The unconditional power of the level- α test that rejects when $F > F_\alpha(q, n - p)$ is $E[\pi(\gamma(\mathbf{X}))]$, where $\pi(\gamma) = \Pr\{F(q, n - p, \gamma) > F_\alpha(q, n - p)\}$.

Proof. Fix a value of $\mathbf{X}$ of rank p . Given this value, $\mathbf{Y}$ follows the normal linear model with a fixed model matrix, so the results of Chapter 7 apply to the conditional law given $\mathbf{X}$.

- (a) By Corollary 7.5.2 with $d = \boldsymbol{\Lambda}^\top \boldsymbol{\beta}$, the conditional law of T is $t(n - p)$. It is the same for every design, so Lemma 14.1.1 gives the unconditional law and the independence.
- (b) Write $\mathbf{W} = \boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\Lambda}$, which is positive definite because $\boldsymbol{\Lambda}$ has rank q . By Theorem 7.5.1(a), given $\mathbf{X}$, $\mathbf{D} = \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d} \sim N_q(\boldsymbol{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d}, \sigma^2 \mathbf{W})$, so by Theorem 4.3.2(b) with $\mathbf{A} = \mathbf{W}^{-1} / \sigma^2$ (for which $\mathbf{A} \cdot \sigma^2 \mathbf{W} = \mathbf{I}$ is idempotent) the quadratic form $\mathbf{D}^\top \mathbf{W}^{-1} \mathbf{D} / \sigma^2$ is $\chi^2(q, \gamma(\mathbf{X}))$. It is independent of $(n - p)s^2 / \sigma^2 \sim \chi^2(n - p)$ by Theorem 7.5.1(b) and (c). Hence the conditional law of F is $F(q, n - p, \gamma(\mathbf{X}))$ (Definition 4.1.2). Under $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ the noncentrality is zero for every design, and Lemma 14.1.1 applies.
- (c) Let A be the event that the set covers the true value, or that the test rejects. Then $\Pr(A) = E[\Pr(A | \mathbf{X})] = E(1 - \alpha) = 1 - \alpha$, and likewise for the size.
- (d) Given $\mathbf{X}$, the rejection probability is $\pi(\gamma(\mathbf{X}))$ by (b). Take expectations.

The statistic in (b) is the F statistic of the general linear hypothesis (Theorem 11.3.2) for a full-rank design. So every test of Chapter 11 and every interval of Chapter 12 that is exact for fixed $\mathbf{X}$ is exact for random $\mathbf{X}$ too, whatever the distribution of the regressors, even without finite moments.

Example. (Skewed regressors)

Take $n = 10$ cases with an intercept and two regressors drawn afresh for every sample, one exponential and one lognormal, and $E(Y) = 1 + 0 \cdot x_1 + 0.5x_2$. The t test of $\beta_1 = 0$ at level 0.05 uses the $t(7)$ quantile. Over 100000 samples with standard normal errors independent of the regressors, the test rejects in a proportion 0.0508 of them, as Theorem 14.1.2 requires; panel (b) of Figure 14.1.1 shows the simulated statistics against the $t(7)$ density. If instead the error of each case is multiplied by its own value of x_1 , so that the spread depends on the regressor, the same test rejects in a proportion 0.2798. The conditional model no longer holds, and nothing in the theorem protects the test.

```
import numpy as np
from scipy import stats
reps = 100_000

rng = np.random.default_rng(1402)
n2, p2 = 10, 3
beta = np.array([1.0, 0.0, 0.5])          # H0: beta_1 = 0 is true
tq = stats.t.ppf(0.975, n2 - p2)

def t_stats(errors_from):
    """t statistics for beta_1 = 0 over many samples with skewed random regressors.
    """
    out = np.empty(reps)
    for s in range(reps):
        X = np.column_stack([np.ones(n2), rng.exponential(size=n2), rng.lognormal(
            size=n2)])
        y = X @ beta + errors_from(X)
        XtX_inv = np.linalg.inv(X.T @ X)
        b = XtX_inv @ X.T @ y
        s2 = np.sum((y - X @ b) ** 2) / (n2 - p2)
        out[s] = b[1] / np.sqrt(s2 * XtX_inv[1, 1])
    return out

t_indep = t_stats(lambda X: rng.normal(size=n2))          # errors independent of X
t_hetero = t_stats(lambda X: X[:, 1] * rng.normal(size=n2)) # spread grows with x_1
for label, t in [("independent errors", t_indep), ("spread depends on X", t_hetero)]:
    print(f"{label:20s} rejection rate {np.mean(np.abs(t) > tq):.4f}")
```

[Run this code online](#)

Warning. The conditional model can fail in ways no amount of data repairs. If the error variance depends on the regressors, the t and F statistics lose their exact laws (Chapter 21). If the regressors are correlated with the errors, as with measurement error (Chapter 24), an omitted common cause (Chapter 25) or a lagged response, least squares estimates something other than β . And if the mean is not linear, the population

projection of [Section 6.11](#) is still estimated consistently, but its errors depend on the regressors and the usual standard errors are wrong.

14.1.3 What changes: the law of the estimates

Pivots keep their laws, but estimates do not. Given the design, $\hat{\beta} \sim N_p(\beta, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1})$. Unconditionally $\hat{\beta}$ is a mixture of normal distributions over the random covariance $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$. Its mean is still β , and its covariance is $\sigma^2 E[(\mathbf{X}^\top \mathbf{X})^{-1}]$ by Corollary 5.5.5, but it is not normal. For a single normal regressor the mixture can be written down exactly. We need one moment of the chi-squared distribution.

Lemma 14.1.3 — Mean of an inverse chi-squared variable

If $V \sim \chi^2(\nu)$ with $\nu > 2$, then $E(1/V) = 1/(\nu - 2)$, the fact used without proof for Eq. 9.6.2. For $\nu \leq 2$ the mean is infinite.

Proof. With the density Eq. 4.1.1,

$$E\left(\frac{1}{V}\right) = \int_0^\infty \frac{v^{\nu/2-2} e^{-v/2}}{2^{\nu/2} \Gamma(\nu/2)} dv = \frac{2^{\nu/2-1} \Gamma(\nu/2 - 1)}{2^{\nu/2} \Gamma(\nu/2)} = \frac{1}{2(\nu/2 - 1)} = \frac{1}{\nu - 2},$$

using $\Gamma(a + 1) = a\Gamma(a)$. The integral converges at zero iff $\nu/2 - 2 > -1$, that is, $\nu > 2$. ■

Proposition 14.1.4 — The slope under a random normal regressor

Let (X_i, Y_i) , $i = 1, \dots, n$ with $n \geq 3$, be independent draws from a bivariate normal distribution with $\text{Var}(X_i) = \sigma_X^2$, regression slope β_1 and conditional variance $\sigma^2 = \text{Var}(Y_i | X_i)$. Let $\hat{\beta}_1$ be the least squares slope. Then

$$\frac{\sigma_X \sqrt{n-1}}{\sigma} (\hat{\beta}_1 - \beta_1) \sim t(n-1),$$

so $\hat{\beta}_1$ has heavier tails than any normal distribution, and for $n > 3$

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sigma_X^2 (n-3)}. \quad (14.1.2)$$

Proof. Given the X values, $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{xx})$ with $S_{xx} = \sum_i (X_i - \bar{X})^2$ (Theorem 7.5.1(a) and Corollary 5.5.4). So $Z = \sqrt{S_{xx}}(\hat{\beta}_1 - \beta_1)/\sigma$ is standard normal given every design, and by [Lemma 14.1.1](#) it is standard normal and independent of the X values, hence of $V = S_{xx}/\sigma_X^2$. The X_i are independent $N(\mu_X, \sigma_X^2)$, so $V \sim \chi^2(n-1)$ by

Corollary 4.4.5(b). Therefore

$$\frac{\sigma_X \sqrt{n-1}}{\sigma} (\hat{\beta}_1 - \beta_1) = \frac{Z}{\sqrt{V/(n-1)}} \sim t(n-1)$$

by Definition 4.1.3. For the variance, $E(Z^2/V) = E(Z^2) E(1/V) = 1/(n-3)$ by independence and Lemma 4.1.3, so $\text{Var}(\hat{\beta}_1) = \sigma^2 \sigma_X^{-2}/(n-3)$. ■

The naive guess, the fixed-design variance σ^2/S_{xx} with S_{xx} replaced by its mean $(n-1)\sigma_X^2$, is too small by the factor $(n-3)/(n-1)$. This is Jensen's inequality at work, as in Example (One slope, two sampling schemes): designs that happen to bunch the regressor values together cost more precision than spread-out designs gain. Section 14.2 extends Eq. 14.1.2 to several regressors.

Example. (A heavy-tailed slope)

With $n = 12$, $\sigma_X = 2$, $\sigma = 1$ and slope 0.5, the variance of the slope over 100000 simulated samples, each with its own design, is 0.02751, against 0.02778 from Eq. 14.1.2 and 0.02273 from the naive guess. The excess kurtosis of the simulated slopes is 0.85, close to the value 0.86 of the $t(11)$ distribution. Panel (a) of Figure 14.1.1 shows the tails on a logarithmic scale. The scaled $t(11)$ density follows the simulated slopes, and the normal density with the same variance falls away too fast.

```
rng = np.random.default_rng(1401)
n, beta1, sigma, sigma_x = 12, 0.5, 1.0, 2.0
x = sigma_x * rng.normal(size=(reps, n))          # a new design for every sample
y = 1.0 + beta1 * x + sigma * rng.normal(size=(reps, n))
xc = x - x.mean(axis=1, keepdims=True)
Sxx = np.sum(xc ** 2, axis=1)
b1 = np.sum(xc * y, axis=1) / Sxx                  # least squares slope, one per
sample

scale = sigma / (sigma_x * np.sqrt(n - 1))          # b1 - beta1 = scale * t(n-1)
print(f"variance of the slope   {b1.var():.5f}")
print(f"sigma^2/(sigma_x^2 (n-3)) {sigma**2 / (sigma_x**2 * (n - 3)):.5f}")
print(f"sigma^2/E(Sxx)           {sigma**2 / (sigma_x**2 * (n - 1)):.5f}")
print("KS p-value against the scaled t(n-1):",
      round(stats.kstest((b1 - beta1) / scale, "t", args=(n - 1,)).pvalue, 3))
```

[Run this code online](#)

Which of the two distributions should a report describe? For the data in hand, the conditional one. The design is *ancillary*: its distribution does not involve (β, σ^2) , and it tells us how precise this particular sample is. A slope estimated from widely spread regressor values deserves a shorter interval than one estimated from bunched values, which the conditional standard error $s/\sqrt{S_{xx}}$ delivers, and by Theorem 14.1.2 the interval is also exact on average over designs. The unconditional law answers questions asked before the data arrive, such as the power of a planned study.

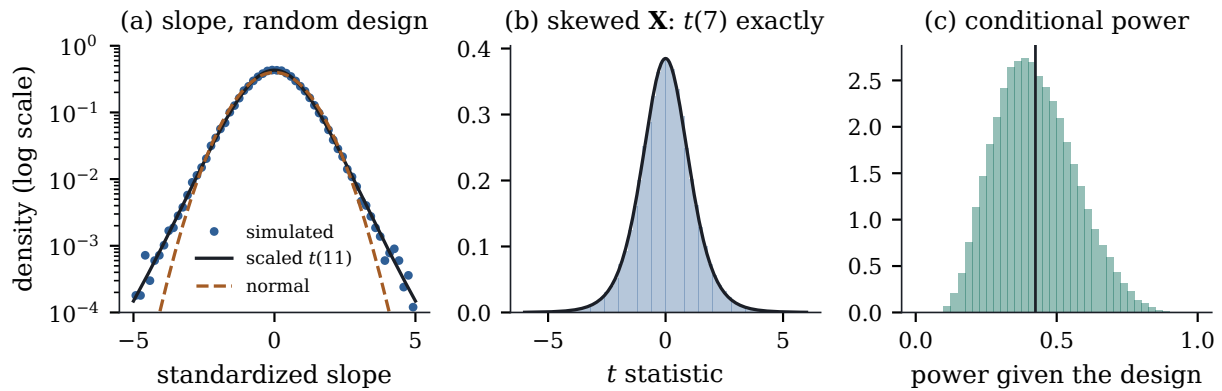


Figure 14.1.1: Random regressors. (a) The standardized slope with a random normal regressor ($n = 12$) follows the scaled $t(11)$ law of Proposition 14.1.4. (b) With skewed regressors the t statistic is exactly $t(7)$. (c) Conditional power varies with the design; the line marks the unconditional power.

14.1.4 Power depends on the design

Part (d) of Theorem 14.1.2 says that power, unlike size, is not free of the design. For a single normal regressor the noncentrality of the test of a zero slope is $\gamma(\mathbf{X}) = \beta_1^2 S_{xx} / \sigma^2$, which is proportional to a $\chi^2(n-1)$ variable.

Example. (How much power has this design?)

Take $n = 12$, $\sigma = \sigma_X = 1$ and slope 0.6. The test of a zero slope at level 0.05 has conditional power between 0.321 and 0.518 for the middle half of the designs, with median 0.414 (panel (c) of Figure 14.1.1). The unconditional power, the average over designs, is 0.425; a direct simulation of the whole procedure gives 0.426. Plugging in the average design, $S_{xx} = n - 1$, gives 0.436, which overstates the power a study should expect.

```
rng = np.random.default_rng(1403)
n3, b3, df3 = 12, 0.6, 10                                # slope 0.6, sigma = sigma_x =
1                                                         1
Sxx3 = stats.chi2.rvs(n3 - 1, size=reps, random_state=rng) # Sxx / sigma_x^2 ~ chi
~2(n-1)
gamma = b3 ** 2 * Sxx3                                     # noncentrality given the
design
fq = stats.f.ppf(0.95, 1, df3)
cond_power = stats.ncf.sf(fq, 1, df3, gamma)               # power of the F = t^2 test
given X
print(f"conditional power: quartiles {np.percentile(cond_power, [25, 50, 75]).round
(3)}")
print(f"unconditional power {cond_power.mean():.3f}")
print(f"power at the average design {stats.ncf.sf(fq, 1, df3, b3**2 * (n3 - 1)):.3f}
")
```

[Run this code online](#)

14.1.5 Exercises

14.1.5.1 A. Check your understanding

Exercise A1.

In the setting of [Proposition 14.1.4](#) take $n = 20$, $\sigma = 2$ and $\sigma_X = 1$. Compute the unconditional variance of $\hat{\beta}_1$ and compare it with $\sigma^2/E(S_{xx})$. Which quantile would you use for a 95% interval for β_1 computed from one sample?

Solution. $\text{Var}(\hat{\beta}_1) = 4/17 = 0.235$, against $4/19 = 0.211$ for the naive value. For an interval from one sample, use the conditional standard error $s/\sqrt{S_{xx}}$ and the $t(18)$ quantile, as for a fixed design; by [Theorem 14.1.2](#) its coverage is exactly 95% over random designs too. The unconditional $t(19)$ law of the proposition involves the unknown σ/σ_X and is not used for intervals.

Exercise A2.

Under [Definition 14.1.1](#), which of the following hold unconditionally: (i) $E(s^2) = \sigma^2$; (ii) $\text{Cov}(\hat{\beta}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$; (iii) $(n-p)s^2/\sigma^2 \sim \chi^2(n-p)$, independent of $\mathbf{X}$; (iv) the 95% confidence ellipsoid for β has coverage 0.95?

14.1.5.2 B. Practice

Exercise B1.

In the setting of [Proposition 14.1.4](#), let x_0 be a fixed value and $\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$ the fitted mean there. Show that

$$\text{Var}(\hat{\mu}_0) = \sigma^2 \left[\frac{1}{n} + \frac{(x_0 - \mu_X)^2 + \sigma_X^2/n}{\sigma_X^2(n-3)} \right].$$

Hint: $\bar{X}$ and S_{xx} are independent.

Solution. Given the design, $\hat{\mu}_0$ is unbiased with variance $\sigma^2[1/n + (x_0 - \bar{X})^2/S_{xx}]$ (Eq. 5.5.2). By [Proposition 2.5.4](#), the unconditional variance is the expectation of this, since the conditional mean $\beta_0 + \beta_1 x_0$ is constant. By [Corollary 4.4.5](#), $\bar{X} \sim N(\mu_X, \sigma_X^2/n)$ is independent of $S_{xx} = \sigma_X^2 V$ with $V \sim \chi^2(n-1)$. Hence $E[(x_0 - \bar{X})^2/S_{xx}] = E(x_0 - \bar{X})^2 E(1/V)/\sigma_X^2 = [(x_0 - \mu_X)^2 + \sigma_X^2/n]/[\sigma_X^2(n-3)]$ by [Lemma 14.1.3](#).

Exercise B2.

Suppose a pilot sample is used to choose the regressor values of the main study, so that the distribution of $\mathbf{X}$ in the main study depends on β . Assume that, given $\mathbf{X}$, the main-study responses follow the normal linear model with errors independent of the pilot. Show that the t test of [Theorem 14.1.2\(a\)](#), computed from the main study alone, still has exact size. Why is the design no longer ancillary, and what information about β does the t test ignore?

14.1.5.3 C. Going deeper**Exercise C1.**

In the setting of [Proposition 14.1.4](#), with σ known, consider the interval $\hat{\beta}_1 \pm 1.96 \sigma / \sqrt{(n-1)\sigma_X^2}$, which uses the average design instead of the observed one. Show that its coverage given the design is $2\Phi(1.96\sqrt{V/(n-1)}) - 1$ with $V \sim \chi^2(n-1)$, and that its unconditional coverage is below 0.95. Which designs make it badly wrong, and in which direction?